
Sniffing---

Basic

With sniffing, you can monitor all sorts of traffic either protected or unprotected with connected network.

The attacker can reveal information from it such as usernames and passwords.

Anyone within the same LAN can sniff the packets.

Working of Sniffers

In the process of sniffing, the attacker gets connected to the target network to start sniffing.

Capture the packet in network

The attacker decrypt the packets to extract information.

Switch vs Hub

- **Switch** switch is a intelligent device beacause it is use mac and ip in packet destination.(send data packet with specific ip)
- **Hub** transmits all packets to all ports.(send data packet with any ip in network)

Spoofing Attacks

MAC Attacks

Media Access Control (MAC) is the physical address of a device. MAC address is a 48-bit unique identification number that is assigned
to a network device for communication at data-link layer (layer 2).

MAC Flooding

Tool - **macof**

python3 macof.py -i <network interface > -s <src ip> -d <dst ip>

python3 macof.py -i eth0 -s 192.168.0.111 -d 192.168.174.130

#Linux#

MAC Changer ***github link = https://github.com/sbdchd/macchanger* :-**

Automatic Method

- ifconfig and check your mac address

- random mac = macchanger -r <interface>

macchanger -r eth0

Manual Method

- ifconfig and check your mac address

- command to change mac adress = macchanger -m <mac address> <interface>

macchanger -m 00-11-22-33-44-55 eth0

Other

-ifconfig (check mac)

-ifconfig eth0 down

-ifconfig eth0 hw ether <mac >

-ifconfig eth0 up

-now check your mac it is changed

#Windows#

Technitium MAC address Changer (Windows) :-

Automatic Mac

- Start Application

- click yes when ask for .tpf (only first time)

- click on "Random MAC Address"
- Now you can see mac is changed

Manual Method

- Start Application
- Select Presets
- Then click New
- Then give Preset Name
- Then click on MAC Address and Go to Use Custom
- And then type Your Target Mac Address
- Then Save
- Then click your Presets Name which you give
- Then click on Apply
- Now your can see Mac is Changed

DHCP Attacks

Dynamic Host Configuration Protocol (DHCP) - DHCP is the process of allocating the IP address dynamically so these addresses are assigned automatically to connected device

- **IPv4 Ports**:
  - UDP port 67 for Server
  - UDP port 68 for Client

- **IPv6 Ports**:
  - UDP port 546 for Client
  - UDP port 547 for Server

ARP Poisoning

Address Resolution Protocol (ARP)

The Address Resolution Protocol (ARP) is a communication protocol

ARP Spoofing Attack

Attacker send forged(fake) ARP packets over Local Area Network (LAN). In this case, switch will update the attacker's MAC address with the IP address

of a legitimate user or server, then start forwarding the packets to the attacker. Attacker can steal information by extracting it from packets. (legitimate

user ki ip ke sath apni(attacker) mac address ko connect ke lena taki wo packet hamer pass aae)

ARP Poisoning used for:

- Session hijacking
- Denial-of-Service attacks
- Man-in-the-Middle attacks
- Packet sniffing
- Data interceptions
- Stealing passwords

Spoofing Attacks

MAC Spoofing/Duplicating

Manipulating the MAC address to impersonate the legitimate user or launch attack such as DoS.

Attacker sniffs the MAC address of users which are active on switch ports and duplicate the MAC address.

This can intercept the traffic and traffic destined to the legitimate user may direct to the attacker.

Wireshark

Filters in Wireshark:

- **`==` Equal**
- **`eq` Equal**
- **`!=` Not equal**
- **`ne` Not equal**
- **ip.src source addresses (Filter ip)**
- **ip.dst destin addresses**
- **ip.addr Match at both the places (source and destin)**

--

Sniffing Countermeasures

- Use Secure Protocol instead of base Protocols (HTTPS over HTTP, SFTP over FTP, etc)

- Switch instead of Hub (Hub broadcast packet by default, but Switch does not)

- Strong encryption protocol (Strong Encrypted data is secure to transmit over any type of network)